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Subject: Data Analytics with R

Experiment No. 01

To Explore and classify various types of datasets with their characteristics and applications in different domains

Title: To explore and classify various types of datasets with their characteristics and applications in different domains

Aim:

The aim of this experiment is to understand, explore, and classify different types of datasets based on their structure, data type, source, and application domain. The experiment also focuses on identifying key characteristics of datasets and suggesting suitable analytics tasks before actual data analysis.

Case Study Overview: Smart City Analytics

Data is gathered from a variety of sources in a smart city setting, including weather stations, hospitals, schools, traffic sensors, and citizen complaint platforms. Understanding each dataset's characteristics is crucial before using any analytical methods. In order to facilitate well-informed decision-making, this experiment conducts an initial investigation and categorization of datasets utilized in a smart city.

Observation Table: Dataset Classification:

Dataset Name	Structure	Data Type	Domain	Key Characteristics	Possible Analytics Task
Bangalore_traffic_Dataset.csv	Structured	Numerical, Time-Series	Traffic Management	Sensor-based, time-dependent, large volume	Traffic congestion prediction
Hospital_Patient_Records.csv	Structured	Numerical, Categorical	Healthcare	Institutional, mixed variables, sensitive data	Disease trend analysis
Students_Performance.csv	Structured	Numerical, Categorical	Education	Academic records, periodic updates	Student performance classification
Telangana_Weather_Data.csv	Structured	Numerical, Time-Series	Weather Forecasting	Continuous sensor data, seasonal trends	Weather trend forecasting
Noise_Complaints.csv	Semi-Structured	Text, Categorical	Public Opinion Analysis	Citizen complaints, subjective, noisy text	Sentiment analysis

Dataset-wise Explanation

1. Bangalore_traffic_Dataset.csv

The traffic conditions gathered from sensors placed throughout Bangalore are represented in this dataset. Numerical, time-dependent data, including vehicle count, traffic density, speed, and timestamps, are included in this structured data. The data clearly demonstrates time dependency because it is gathered on a regular basis. The dataset is primarily used in traffic management systems to examine peak-hour traffic behavior and congestion patterns.

2. Hospital_Patient_Records.csv

This dataset has institutional healthcare records that have both numbers and categories, like age, gender, diagnosis, and treatment details. It is structured and based on records, with each row showing a patient record. When data is entered by hand, it can lead to problems with data quality, like missing or inconsistent values. The dataset enables healthcare analytics, including disease trend analysis and patient classification.

3. Students_Performance.csv

This dataset contains information about students' academic performance that schools have gathered. It has structured numbers, like grades and attendance, as well as categorical data, like gender or course. The data is updated regularly, but it may have missing values because people don't show up. This dataset is helpful for figuring out what affects student performance and for looking at how well students are doing in school.

4. Telangana_Weather_Data.csv

This dataset consists of weather parameters such as temperature, humidity, rainfall, and wind speed collected over time from weather stations. It is structured and shows strong time dependency, making it suitable for time-series analysis. Sensor-related issues may introduce noise or missing values. The dataset supports weather forecasting and climate trend analysis.

5. Noise_Complaints.csv

This dataset contains noise-related complaints submitted by citizens through public platforms. While fields like date, location, and complaint category are structured, the complaint descriptions are in free-text format, making the dataset semi-structured. The data may include informal language and subjective expressions, requiring preprocessing before analysis. It is mainly used for sentiment analysis and public opinion assessment.

Observations

1. Most of the datasets used in the smart city project are structured, which means that data analytics tools can easily read and analyze them.
2. Because the data is collected from sensors at regular intervals, traffic and weather datasets show patterns over time.
3. Institutional datasets, such as hospital and student records, include both numbers and categories, which may need some cleaning.
4. The noise complaints dataset is semi-structured because it has both structured fields and free-text descriptions.
5. The type of analysis that can be done, like prediction, classification, or sentiment analysis, depends on the type of dataset.

Conclusion

This experiment looked at different smart city datasets and sorted them into groups based on their structure, data type, source, and application domain. The study elucidated the methodologies employed in the collection of real-world data from sensors, institutions, and citizens. It was noted that structured datasets facilitate analysis, whereas semi-structured data necessitates further preprocessing. The experiment also showed how important time-series data is for studying traffic and weather. It became clear how to use data analytics in real life by mapping each dataset to its domain. This experiment gave us a good base to do more data analysis with R. In general, knowing the characteristics of a dataset before analyzing it helps you make better and more accurate decisions.